

Fatty Liver

Alcoholic Liver Disease

MASLD



Prof. Wai-Kay Seto

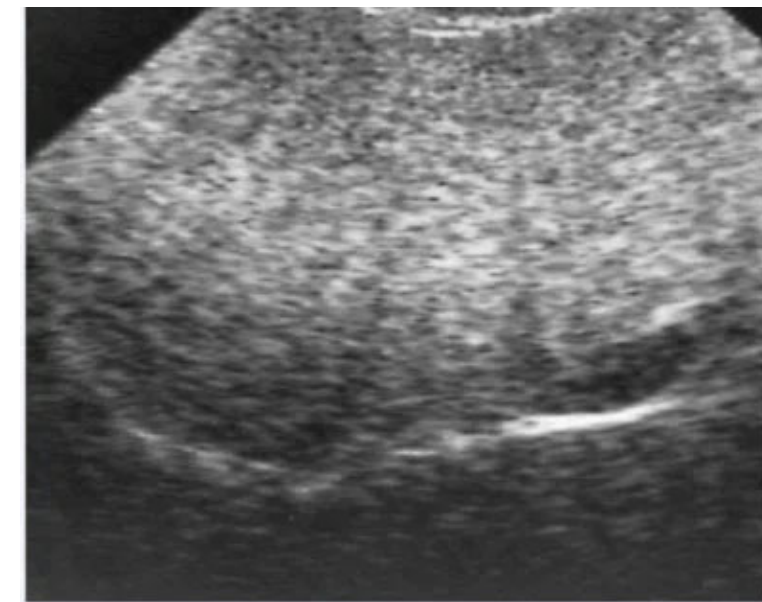
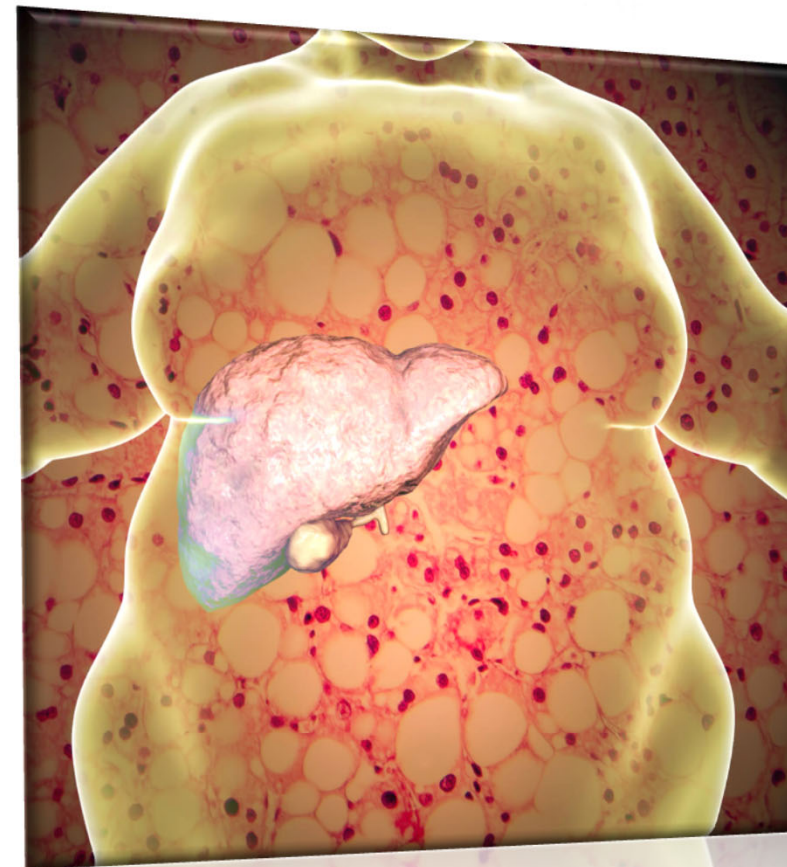
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WCS Outline

- Steatotic Liver Disease
- Alcohol and the Liver
 - Alcoholic fatty liver
 - Alcoholic hepatitis
- Metabolic associated steatotic liver disease (MASLD)
 - MASLD and MASH
 - Management

Related WCSs

- **Cirrhosis & ascites**
- **Liver failure**
- **I am overweight**
- **Alcohol & the brain**

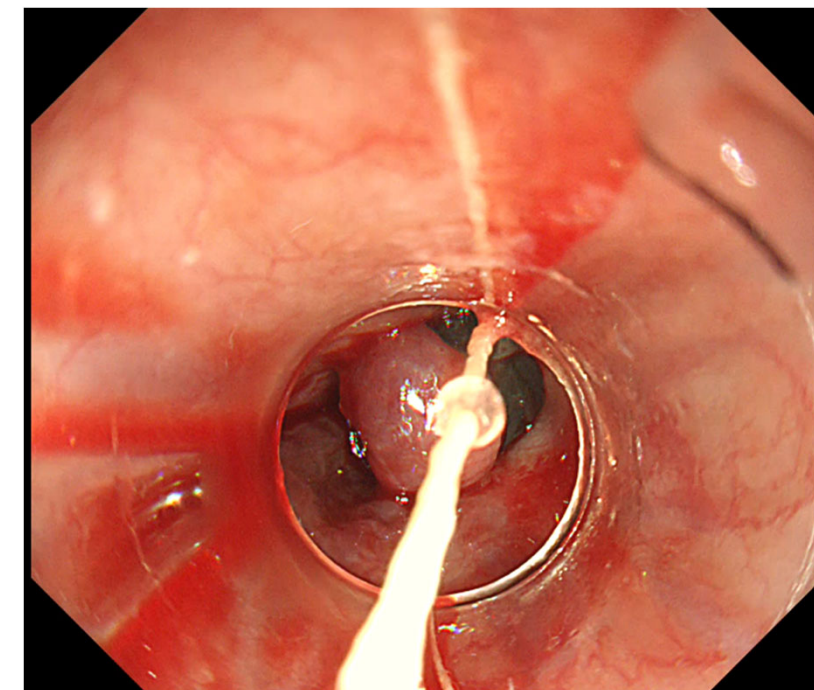
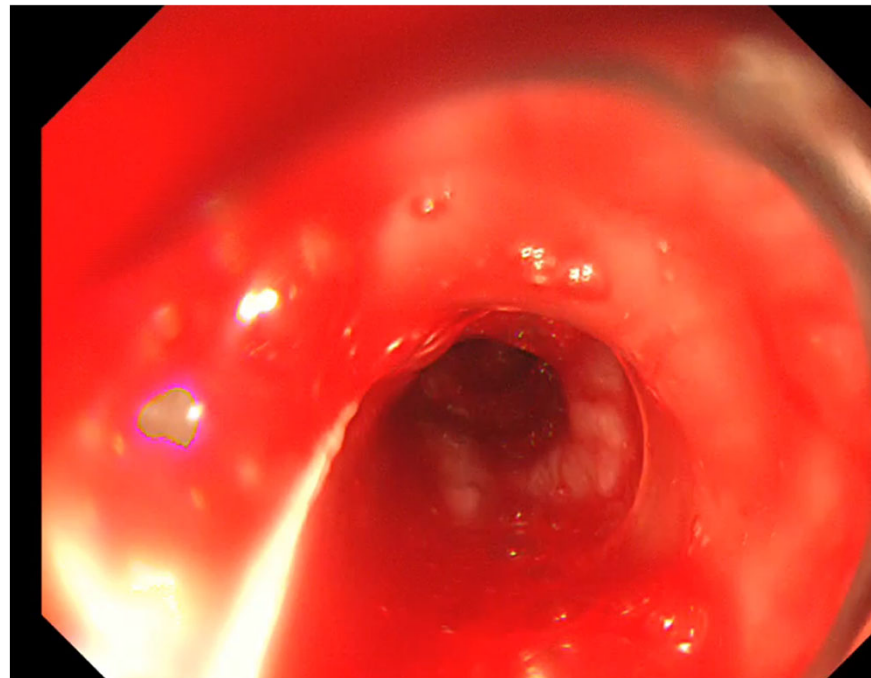
Case Presentation #1

- Male 58 years, work in medical laboratory
- Claimed good health, never need medical consultation
- Noted by colleagues to be jaundiced
- T 37.5C
- USG: no biliary pathology. Splenomegaly+
- Admit to drinking 1 bottle of whisky per day for past 10+ years

Bilirubin	145 umol/L
ALP	116 U/L
AST	278 U/L
ALT	191 U/L
GGT	1155 U/L
INR	1.9
Amylase	214 mmol/L

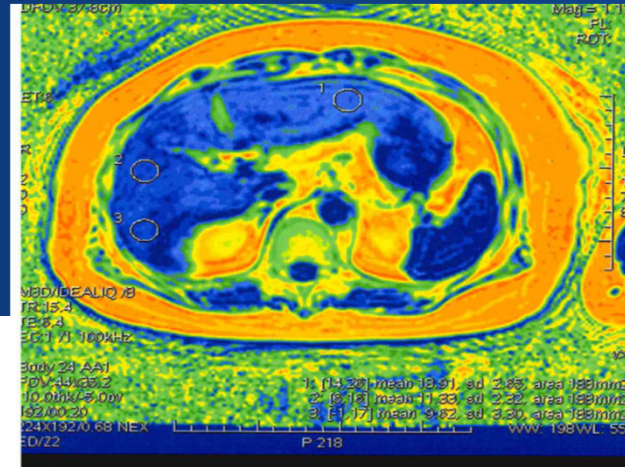
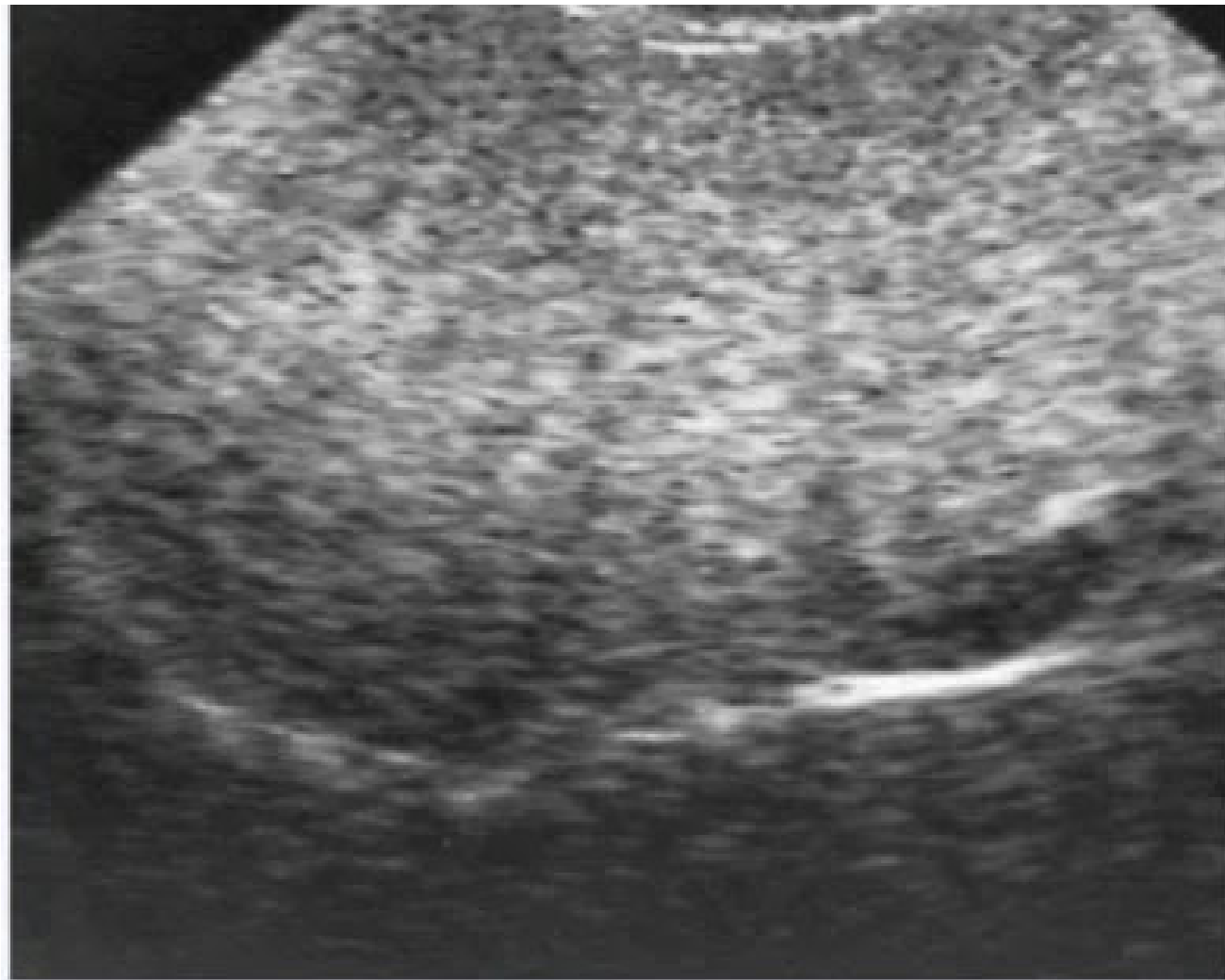
Case Presentation #1

- Seen psychiatry and agreed to alcohol abstinence
- Moderate withdrawal symptoms needing 5 days of lorazepam
- Given iv thiamine and nutritional support
- LFT improved D5 Bilirubin 77 umol/L AST 148 U/L INR 1.4
- D6: Hemetemesis in ward
- OGD: Esophageal variceal bleeding needing banding
- Discharged at D10



Fat in the Liver

Ultrasound



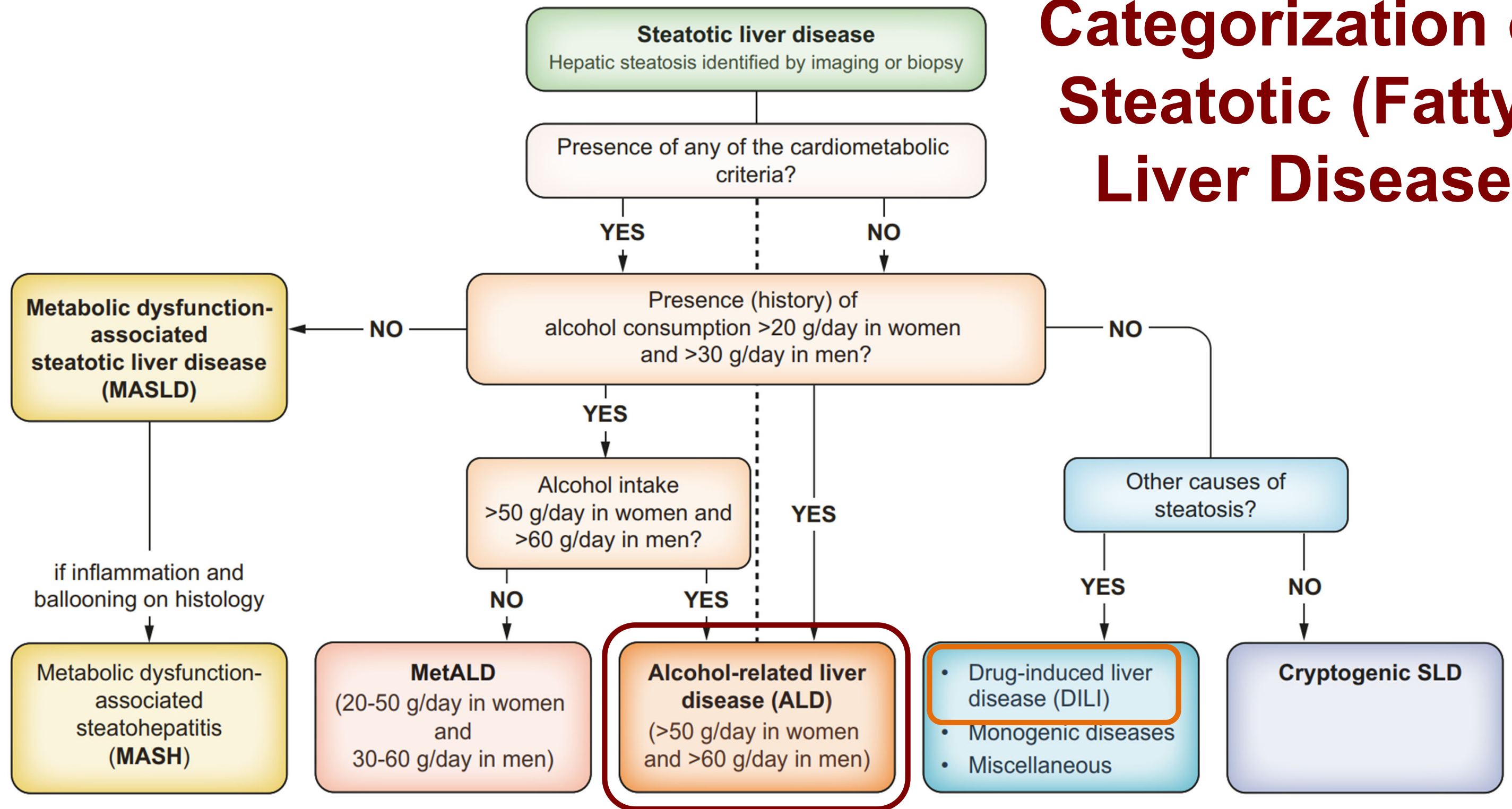
MRI
(specific sequence)

Liver elastography



Steatosis ≥ 248 dB/m
Severe steatosis ≥ 280 dB/m

Categorization of Steatotic (Fatty) Liver Disease



Assessment: Liver Steatosis

- **Detailed** drinking history
- **Cardiometabolic** risk factors
 - Drugs (corticosteroids / tamoxifen / valproate / methotrexate etc)
 - Concomitant liver disease
 - Others (hypothyroid / PCOS / panhypopituitarism / bypass surgery / genetic)
 - LFTs (ALT / AST)
 - Platelet count! (to be explained later)

Every Clinician Needs to Know Alcohol

Drinking is another matter

One Alcoholic Unit = Standard Drink = 10 g Ethanol

- 250 ml beer (5%)
- 100 ml red wine (12%)
- 30 ml spirits (40%)
- And others...

Alcohol Conversion 1 alcohol unit = 10 g pure alcohol

											
0.7 unit	1.3 units	1.3 units	2.0 units	2.5 units	1.2 units	1.3 units	0.7 unit	5.7 units	0.6 unit	0.8 unit	0.3 unit
1 glass (180ml) Beer / alcopops	1 can (330ml) Beer / alcopops	1 bottle (330ml) Beer / alcopops	1 can (500ml) Beer / alcopops	1 bottle (640ml) Beer / alcopops	1 glass (125ml) Red / White wine / Champagne	1 peg (40-50ml) Spirits	1 shot (22ml) Spirits	1 glass (180ml) Rice wine	1 small glass (20ml) Rice wine	1 maotai glass (20ml) Maotai	1 small cup (20ml) Japanese Sake

* Alcohol content is printed on the label of the container.



Soju
16-20%



Sake
15-22%



Sangria
5->15%



Baijiu
40-58%

Alcohol: Global trends

$10\text{ g} = 12.6\text{ mL}$



Global status report
on alcohol and health and treatment
of substance use disorders



Global Alcohol Consumption per Capita

2005
5.5 L

2016
6.4 L

2030
7.6 L
**Western
Pacific**
9.9 L

Socioeconomic status as an effect modifier of alcohol
consumption and harm: analysis of linked cohort data

Srinivasa Vittal Katikireddi, Elise Whitley, Jim Lewsey, Lindsay Gray, Alastair H Leyland

Articles



- Economic status not a prerequisite
- Disadvantaged social groups have greater alcohol-attributable harms

Katikireddi SV et al. Lancet Public Health 2017

Drinking History

AUDIT questionnaire

ALCOHOL AND HEALTH QUESTIONNAIRE (AUDIT#)

Introduction

Because alcohol intake can affect health and interfere with many medications, we want to ask you some questions about your alcohol use. Your answer will remain confidential, so please be as accurate as possible. Please ask for clarification if necessary.

Part A (AUDIT-C) Please circle the answer wherever appropriate.

In the past year...	Scoring system					Your Score
	0	1	2	3	4	
1. How often do you have a drink containing alcohol?	Never	Monthly or less	2 - 4 times per month	2 - 3 times per week	4+ times per week	
2. How many units of alcohol do you drink on a typical day when you are drinking? <i>(With reference to the measurements in the picture below; please add up all types of alcoholic drinks)</i>	0 - 2	3 - 4	5 - 6	7 - 9	10+	
3. How often do you have at least 5 cans of beer/5 glasses of table wine/5 peg of spirits on one occasion? <i>(With reference to the picture below)</i>	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	

SCORE (A): _____

Please turn over if your score is 3 or more

How to Interpret the Score of AUDIT-C?

0	Keep it up! No drinking is a healthy start to a healthy life.
1- 2	- Try to drink less or abstain for preventing cancer and other diseases. - If you drink, limit to 2 alcohol units for men and 1 for women a day. - Complete abstinence if you have medical conditions, are taking medications, operating a vehicle or machinery, are pregnant, or are underage.
3 or above	Please continue to answer the next 7 questions to find out more information (see overleaf).

Q3 scores ≥ 1

You tend to **binge drink** (i.e. consumption of 5 or more alcoholic drinks in a row), which is dangerous and harmful. Try to limit to 2 alcohol units for men and 1 for women a day.

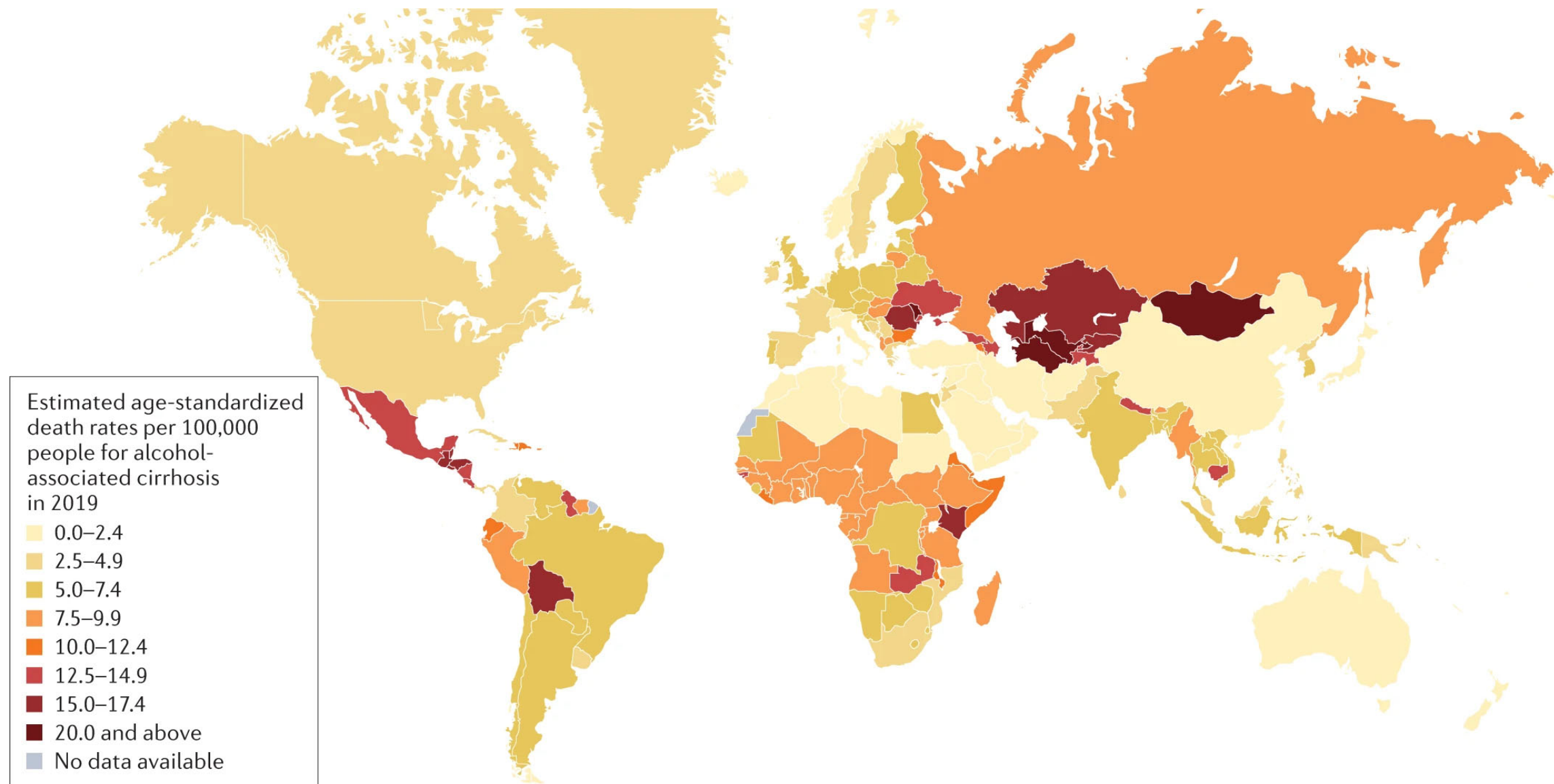
Please turn over if your score is 3 or more

Part B (AUDIT) Please continue to circle the answer wherever appropriate.

(After completing first 3 questions) In the past year...	Scoring system					Your Score
	0	1	2	3	4	
4. How often during the last year have you found that you were not able to stop drinking once you had started?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	
5. How often during the last year have you failed to do what was normally expected from you because of your drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	
6. How often during the last year have you needed an alcoholic drink in the morning to get yourself going after a heavy drinking session?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	
7. How often during the last year have you had a feeling of guilt or remorse after drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	
8. How often during the last year have you been unable to remember what happened the night before because you had been drinking?	Never	Less than monthly	Monthly	Weekly	Daily or almost daily	
9. Have you or somebody else been injured as a result of your drinking?	No		Yes, but not in the last year		Yes, during the last year	
10. Has a relative or friend, doctor or other health worker been concerned about your drinking or suggested that you cut down?	No		Yes, but not in the last year		Yes, during the last year	



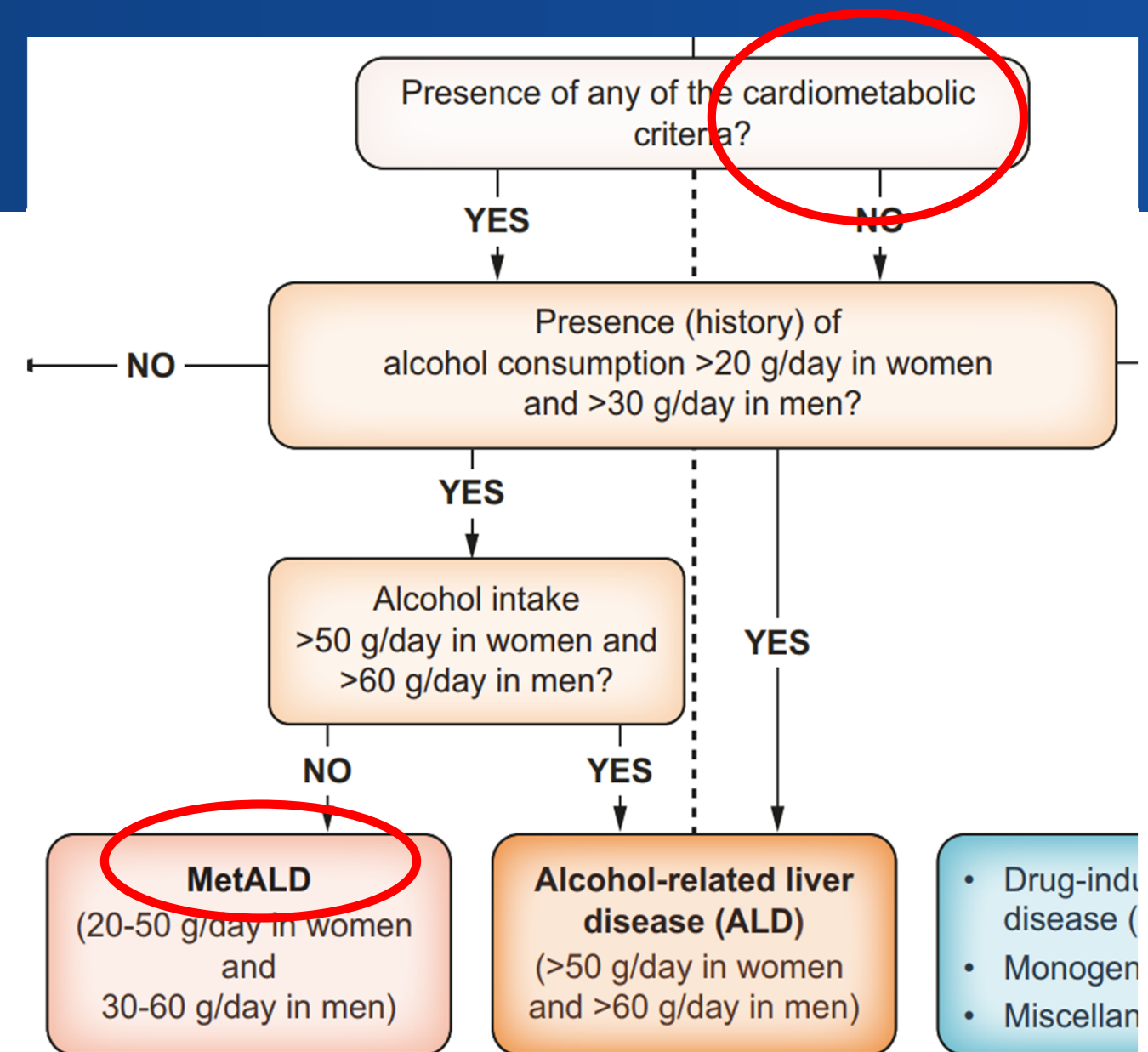
Alcoholic liver cirrhosis and HCC



25% of global cirrhosis
19% of global HCC

Alcohol and the Liver

- **Safety level:** controversial
 - Men 20-30g/day; women 10-20 g/day
 - Binge drinking harmful
- If drinking **50-60 g/day**
 - Alcoholic steatosis: 2+ years
 - Alcoholic hepatitis: 5+ years
 - Alcoholic liver cirrhosis: 5-20 years
- Only gross estimations (drinking history may not be reliable)
- Concomitant liver disease & disease progression (metabolic, viral)



Biomarker for Alcohol Consumption

- **GGT** (can be $>1,000$ U/L) : not specific
- **AST** $>$ ALT during hepatitis: (<400 U/L)
- Serum ethanol: half-life 4-5 hours
- **Phosphatidylethanol (PEth)**
 - Sensitive and specific to ethanol
 - Detection window: 3-5 weeks
 - *Check short-term abstinence*
 - Detectable (>20 mcg/L) with modest drinking
 - >200 mcg/L: heavy drinking

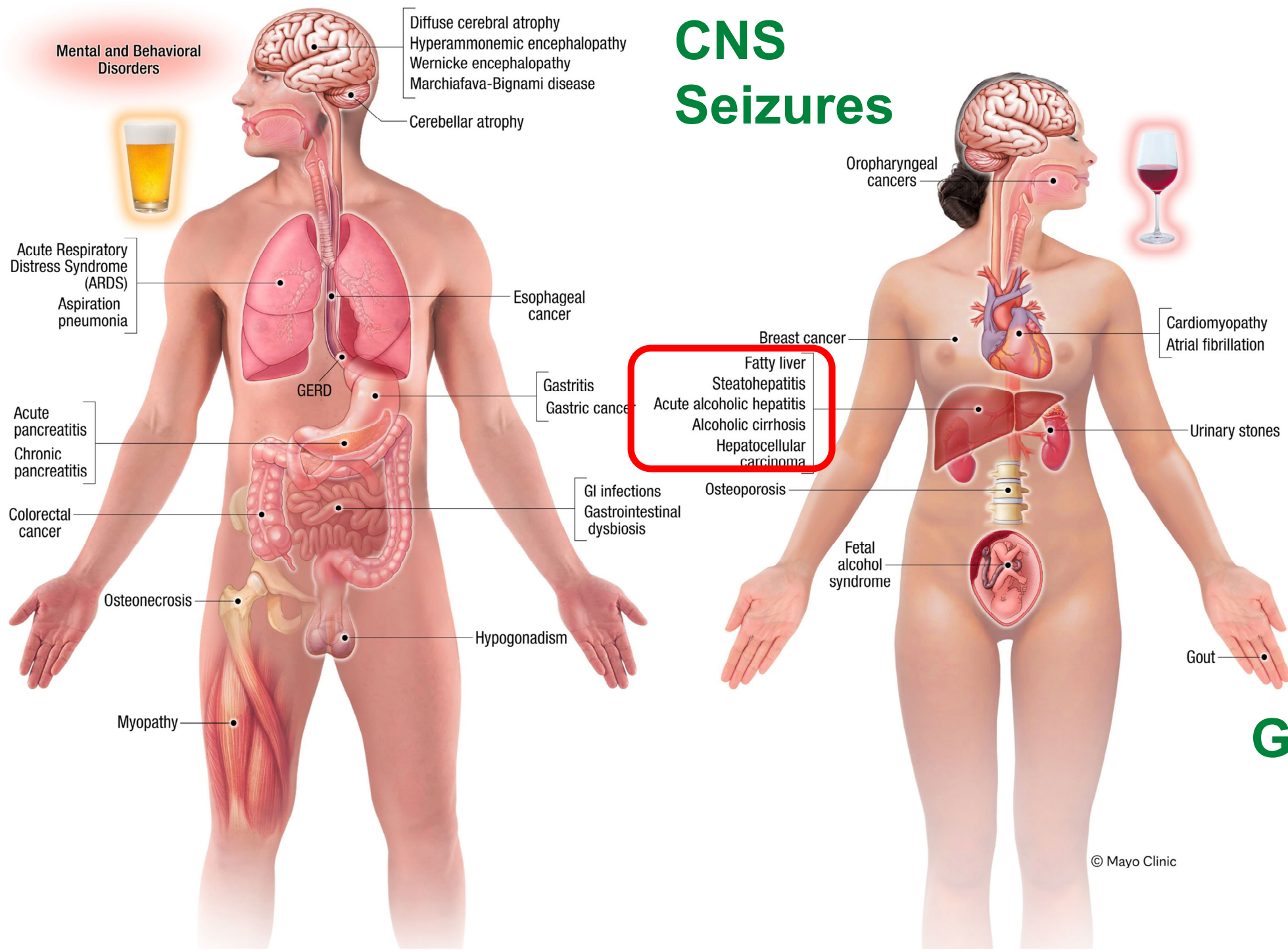


Alcohol: Multiorgan Involvement

Multiple simultaneous presentations

Psychiatric

Acute
Pancreatitis
Chronic
Pancreatitis



CNS
Seizures

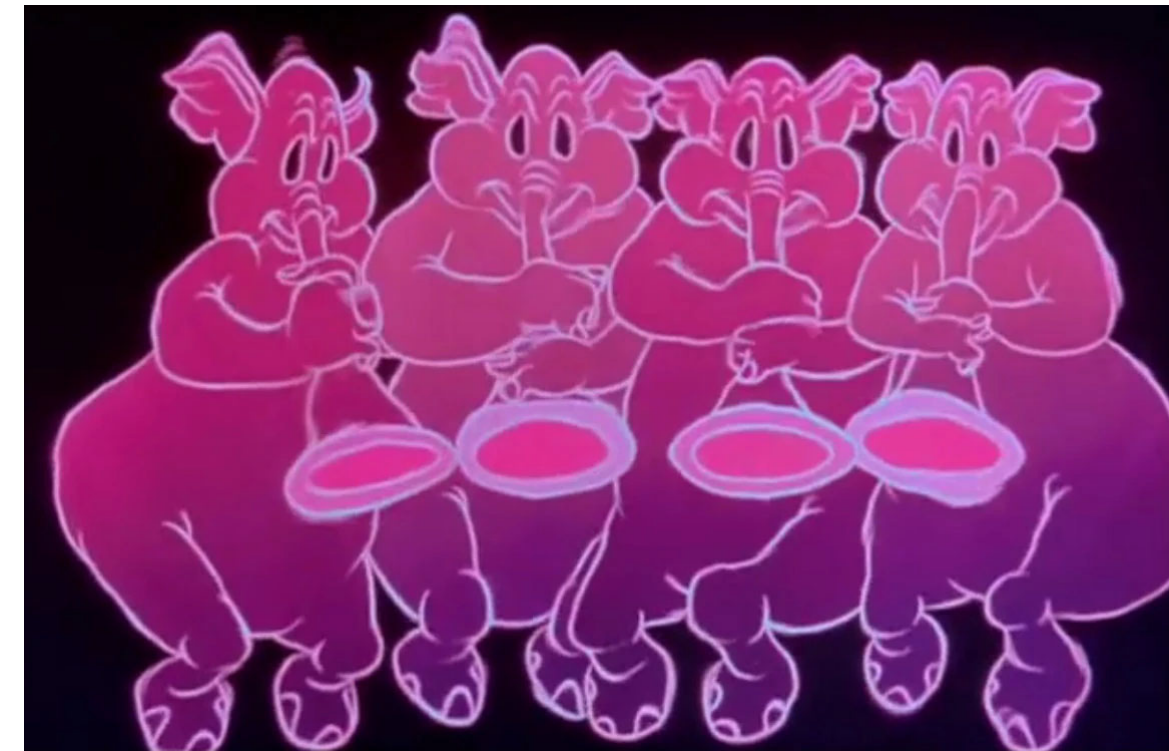
Cardiomyopathy
AF

Gout

© Mayo Clinic

Alcoholic liver disease: Concomitant presentations

- Other organ involvement
 - Psychiatric / CNS / cardiac / pancreatic / others
- Poor nutritional status
- Alcoholic withdrawal (D2+ onward)
 - *Delirium tremens*
- Infections
- Cirrhotic complications
 - *Variceal bleeding / ascites / SBP / HE etc.*



Alcoholic Hepatitis

- Prolonged heavy drinking (>50-60 g/day)
 - *May cut down recently due to symptoms*
- AST:ALT ratio of $\geq 1.5:1$ (AST seldom >400 U/L)
- May have hepatic decompensation ($\uparrow$ INR, Bilirubin)
- +/- abx discomfort / jaundice / low-grade fever
- +/- hemolysis (Zieve's syndrome)
- Exclude:
 - Biliary obstruction (imaging)
 - Other types of hepatitis

**+/- Concomitant
Cirrhotic Complications**

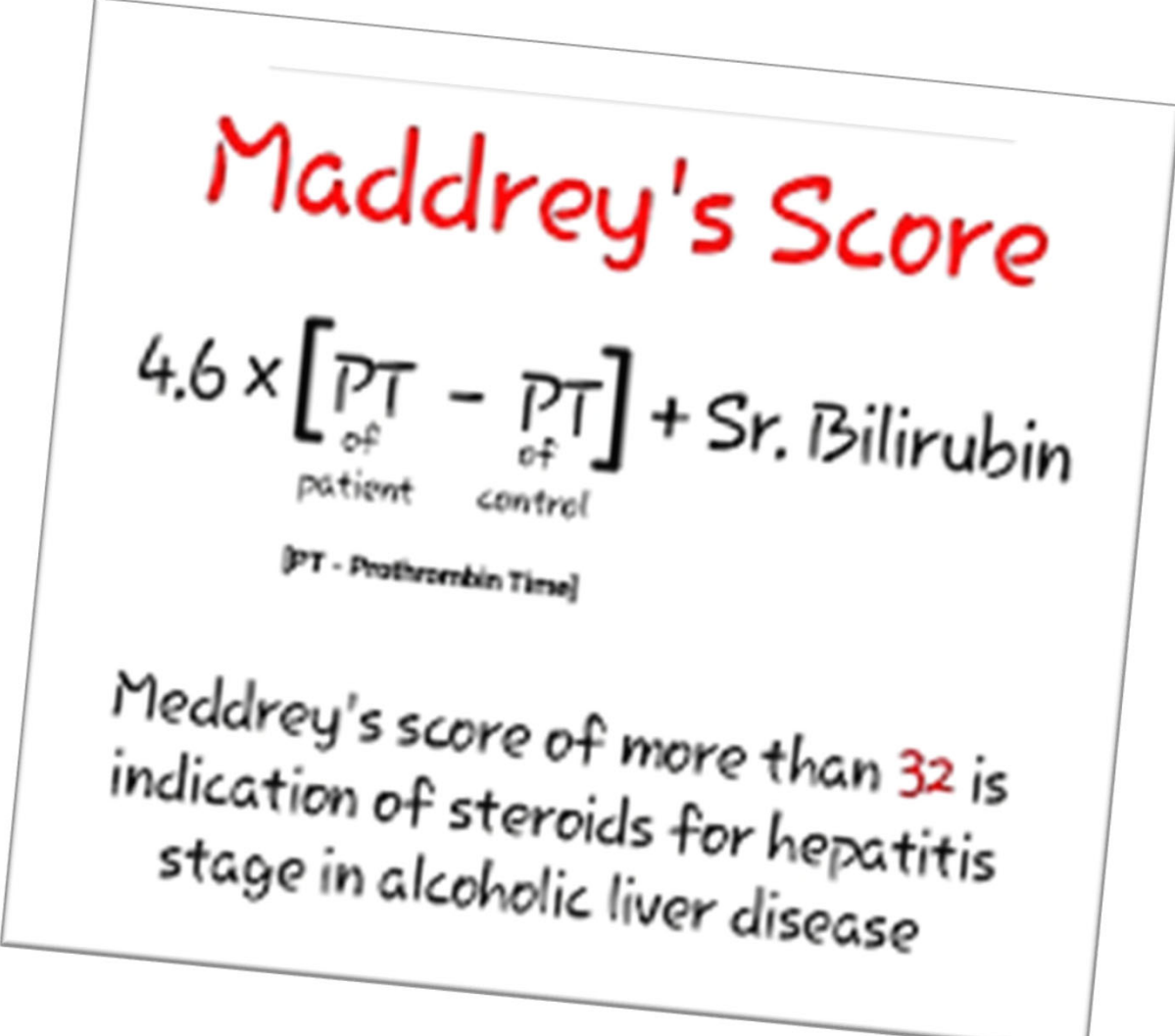
Alcoholic Hepatitis: Determining Severity

- **Maddrey's Discriminant Function**

- Prothrombin time / bilirubin
- ≥ 32 = Severe

- **MELD Score**

- INR / Bilirubin / Creatinine
- ≥ 21 = Severe



Maddrey's Score

$$4.6 \times \left[\underset{\text{patient}}{\text{PT}} - \underset{\text{control}}{\text{PT}} \right] + \text{Sr. Bilirubin}$$

(PT - Prothrombin Time)

Maddrey's score of more than **32** is indication of steroids for hepatitis stage in alcoholic liver disease

Alcoholic Hepatitis: General Management

- Alcohol abstinence! (**Check!!**)
- Supportive (Very important!)
 - Nutritional: Vitamin and mineral replacement
 - *Thiamine*
 - *Folate / PO4 / Mg++*
 - Caloric support. Protein intake 1.2-1.5 g/kg
 - Correct electrolytes

Alcoholic Hepatitis: General Management

- Assess **infection** risk
 - Workup and treat
- Manage other liver-related complications (if present)
 - Variceal bleeding / Ascites / SBP / HE
- Assess and manage other acute alcohol-related conditions
 - Encephalopathy / **Delirium tremens** / Psychiatric
 - Pancreatitis
- Assess for acute kidney injury
- Assess need for ICU support (may be difficult....)

Severe Alcoholic Hepatitis: Pharmaceutical Options

Maddrey > 32 or MELD > 21

- N-acetylcysteine iv infusion
 - Antioxidant
 - Only modest effect
- **Glucocorticoids** (Prednisolone 40mg daily)
 - Improved mortality
 - Difficult decision (risk of infection / GI bleeding etc)
 - Stop if no treatment response by D4
 - Treatment response: gradual taper
- **Liver transplantation?**

Liver Transplantation for Alcohol Liver Disease

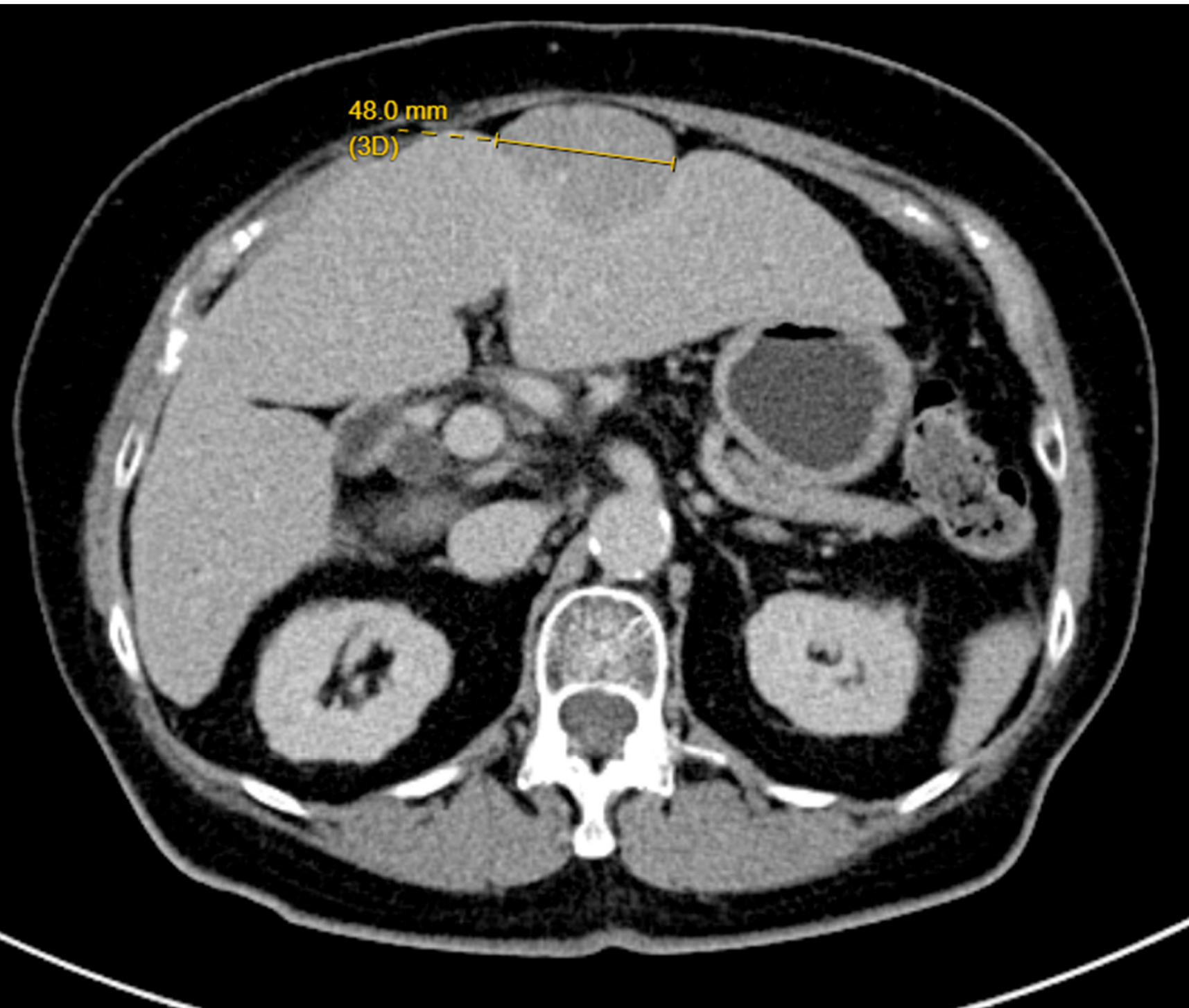
- Deceased Donor:
 - Contraindicated in HK if **last drink $\geq 6M$**
 - Other regions: controversial
- Living Donor:
 - Can be considered if last drink $\geq 6M$
 - Individualized
- Need comprehensive psychosocial assessment
- Issue of alcoholic relapse
- Need to demonstrate commitment to abstinence



Case Presentation #2: a real-world case

- Female 74 years
- BMI 28.6 kg/m²
- Longstanding type 2 DM on insulin, poorly controlled, A1c 9.1%
- CKD (eGFR 30)
- Hypertension / dyslipidemia / minor CAD on 3 anti-HTs / atorvastatin / aspirin
- Regular diabetes complication screening
- Epigastric discomfort, epigastric mass palpated at GOPD

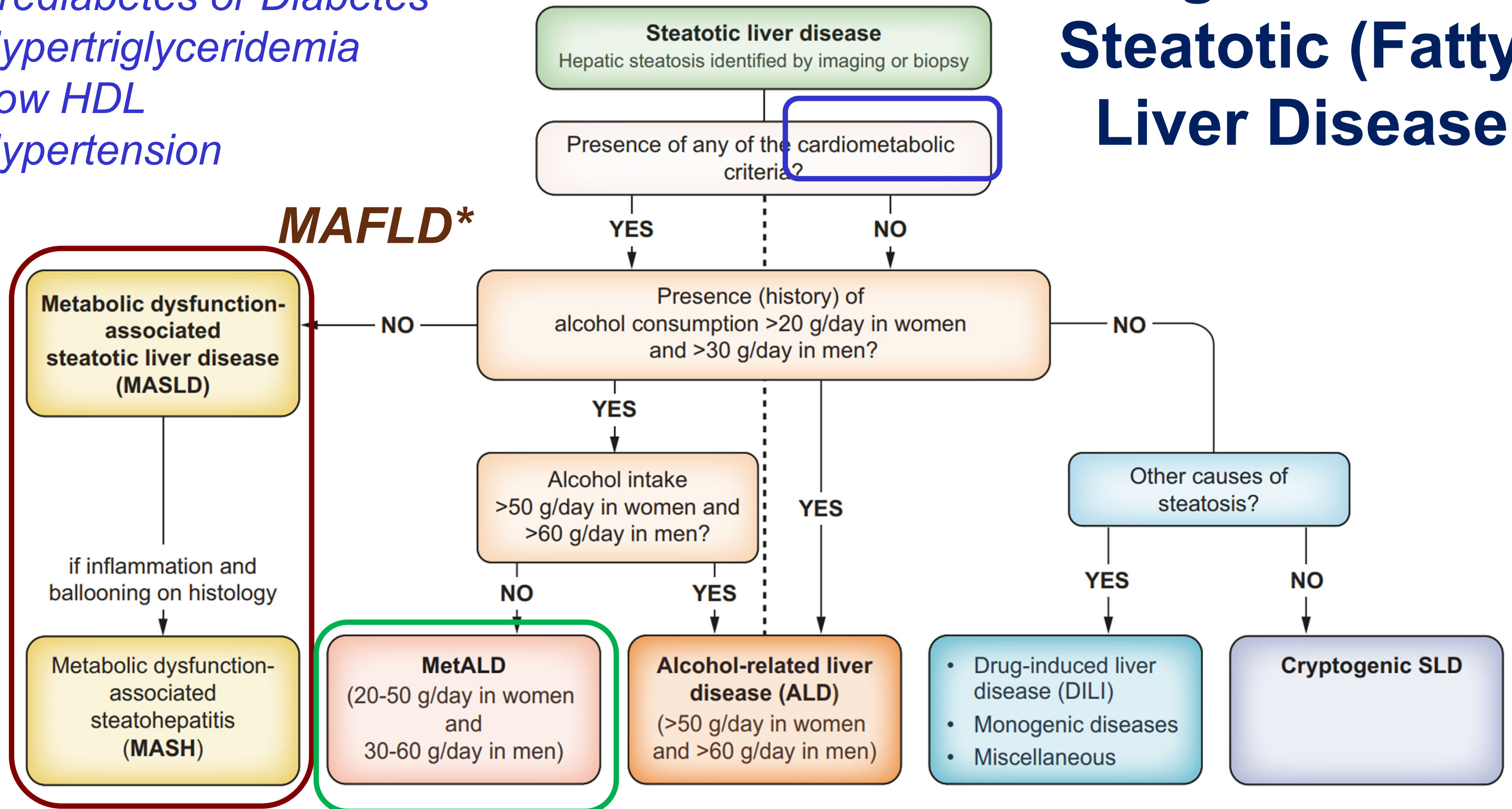
F/74: CT Abdomen



- 4.8 cm S3/S4b HCC
- HBsAg- Anti-HCV- Anti-HBc-
- Slightly lobular liver surface, probable cirrhosis
- Recurrence after left hepatectomy
- Progressive disease with tyrosine kinase inhibitors
- Died 7 months after diagnosis

- *Overweight or obese*
- *Prediabetes or Diabetes*
- *Hypertriglyceridemia*
- *Low HDL*
- *Hypertension*

Categorization of Steatotic (Fatty) Liver Disease



Cardiometabolic Parameter of MASLD

Adult criteria

Body mass index
 $\geq 25 \text{ kg/m}^2$ ($\geq 23 \text{ kg/m}^2$ in people of Asian ethnicity)

Waist circumference

- $\geq 94 \text{ cm}$ in men and $\geq 80 \text{ cm}$ in women (Europeans)
- $\geq 90 \text{ cm}$ in men and $\geq 80 \text{ cm}$ in women (South Asians and Chinese)
- $\geq 85 \text{ cm}$ in men and $\geq 90 \text{ cm}$ in women (Japanese)

Prediabetes: HbA_{1c} 39-47 mmol/mol (5.7-6.4%) or fasting plasma glucose 5.6-6.9 mmol/L (100-125 mg/dl) or 2-h plasma glucose during OGTT 7.8-11 mmol/L (140-199 mg/dl) or

Type 2 diabetes: $\text{HbA}_{1c} \geq 48 \text{ mmol/mol}$ ($\geq 6.5\%$) or fasting plasma glucose $\geq 7.0 \text{ mmol/L}$ ($\geq 126 \text{ mg/dl}$) or 2-h plasma glucose during OGTT $\geq 11.1 \text{ mmol/L}$ ($\geq 200 \text{ mg/dl}$) or

Treatment for type 2 diabetes

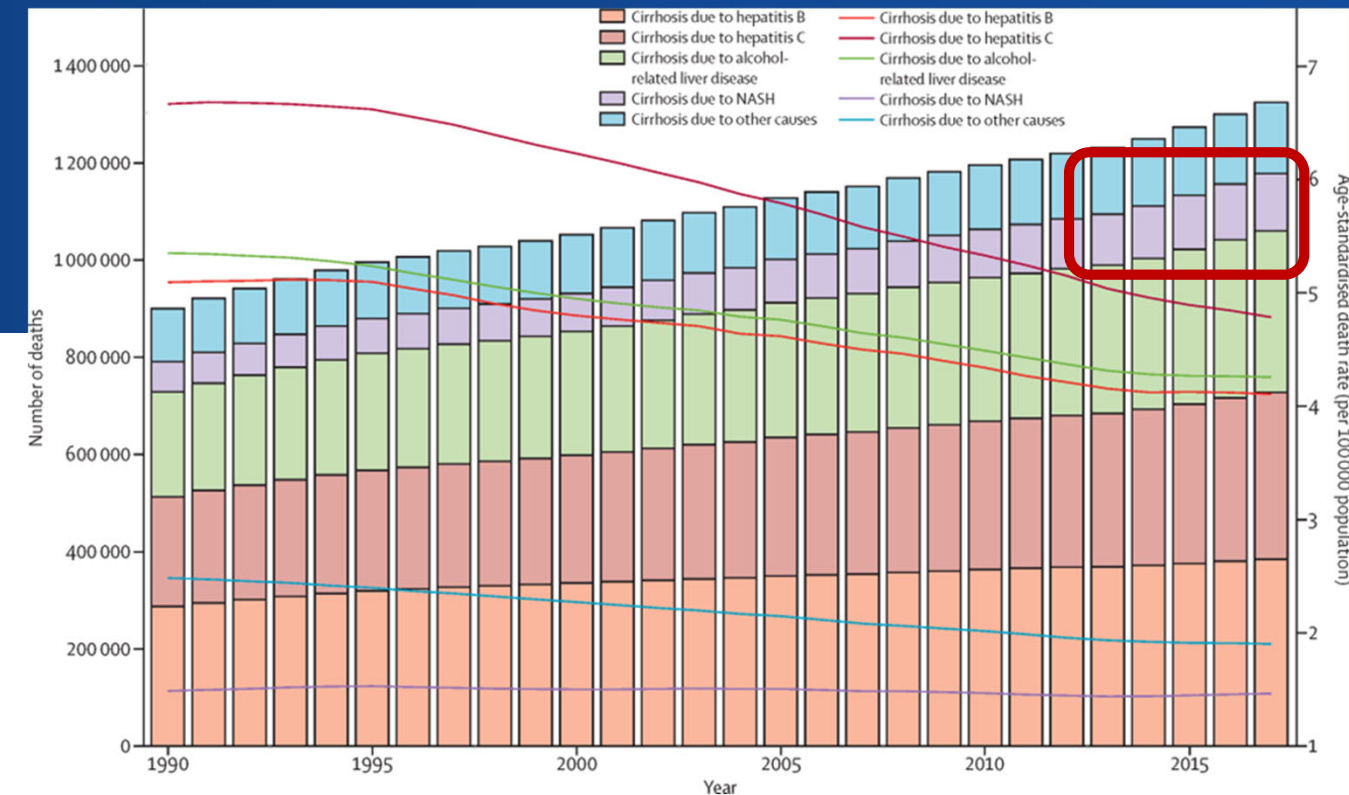
$\geq 1.7 \text{ mmol/L}$ ($\geq 150 \text{ mg/dl}$) or lipid-lowering treatment

$\leq 1.0 \text{ mmol/L}$ ($\leq 39 \text{ mg/dl}$) in men and $\leq 1.3 \text{ mmol/L}$ ($\leq 50 \text{ mg/dl}$) in women or lipid-lowering treatment

$\geq 130/85 \text{ mmHg}$ or treatment for hypertension

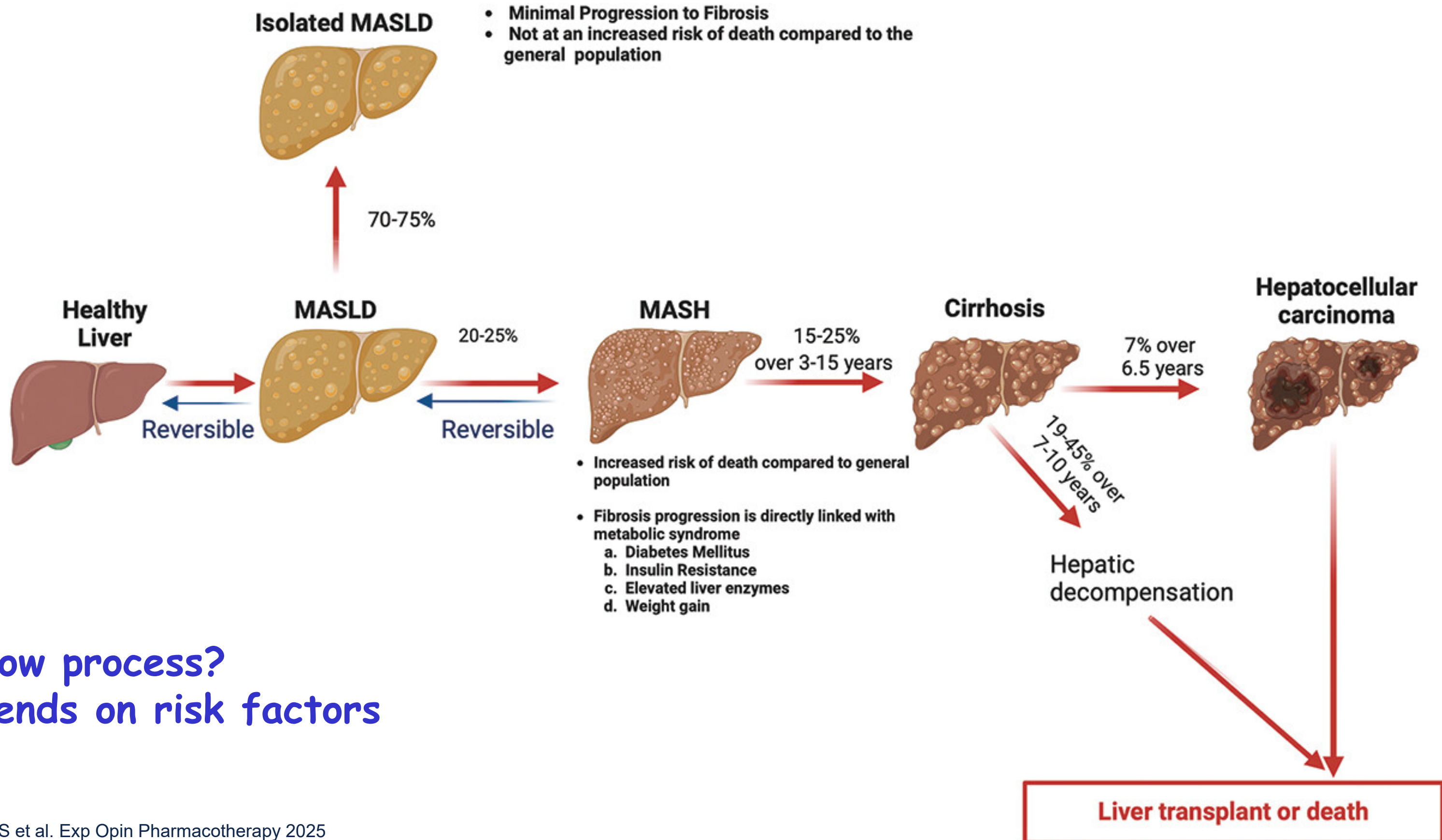
Epidemiology of MASLD

- 30-35% in Asia
 - Main risk factor:
 - Obesity or overweight → → 65%
 - Type 2 DM → → 65-70%
 - 20% with MASH
 - (metabolic associated steatohepatitis)
 - Advanced fibrosis (F3): ~5-8%
 - Cirrhosis: ~3%
- } Higher in DM / Obese



Sole liver disease:
11% of cirrhosis
12% of HCC

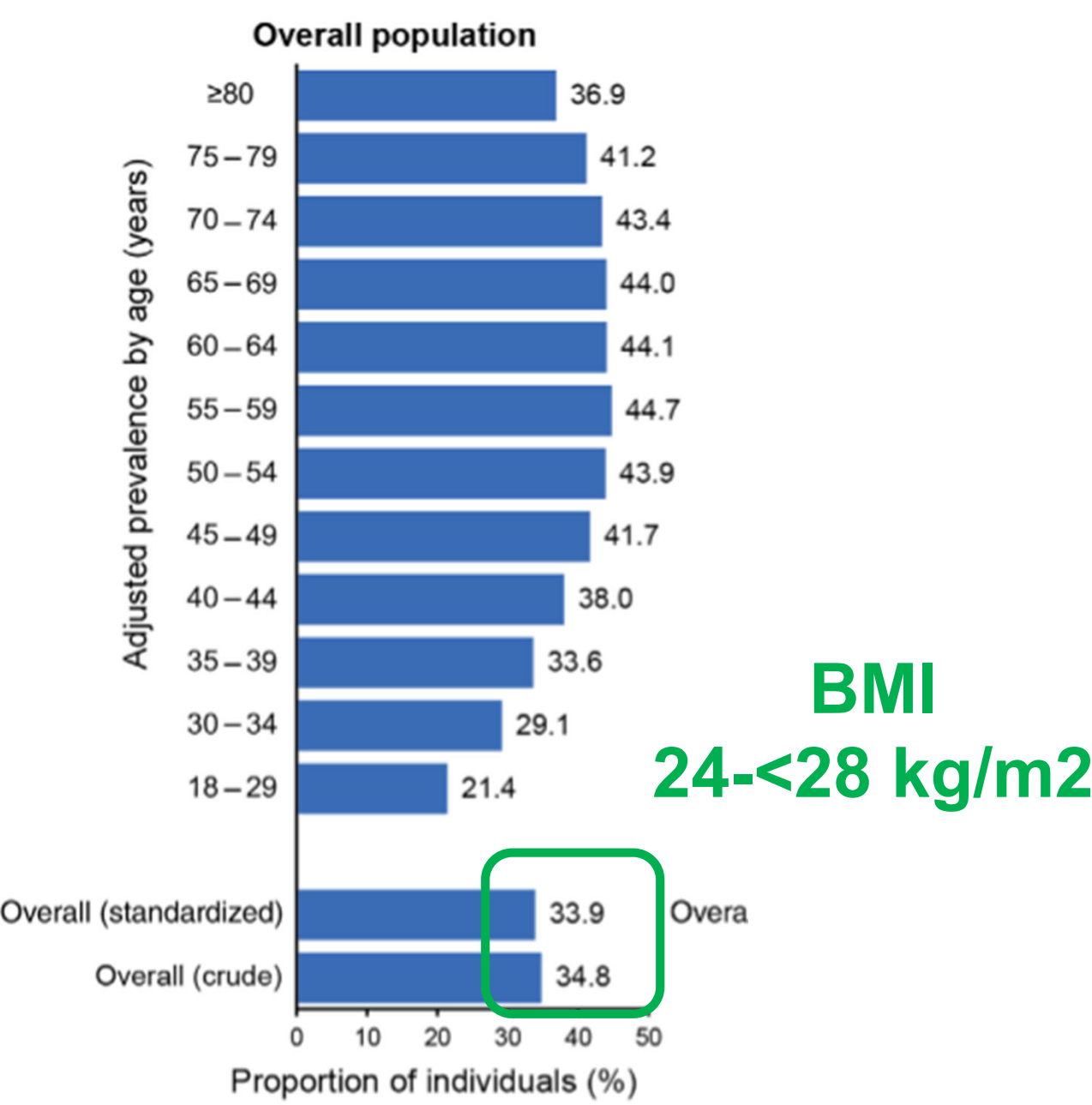
Much higher
concomitantly



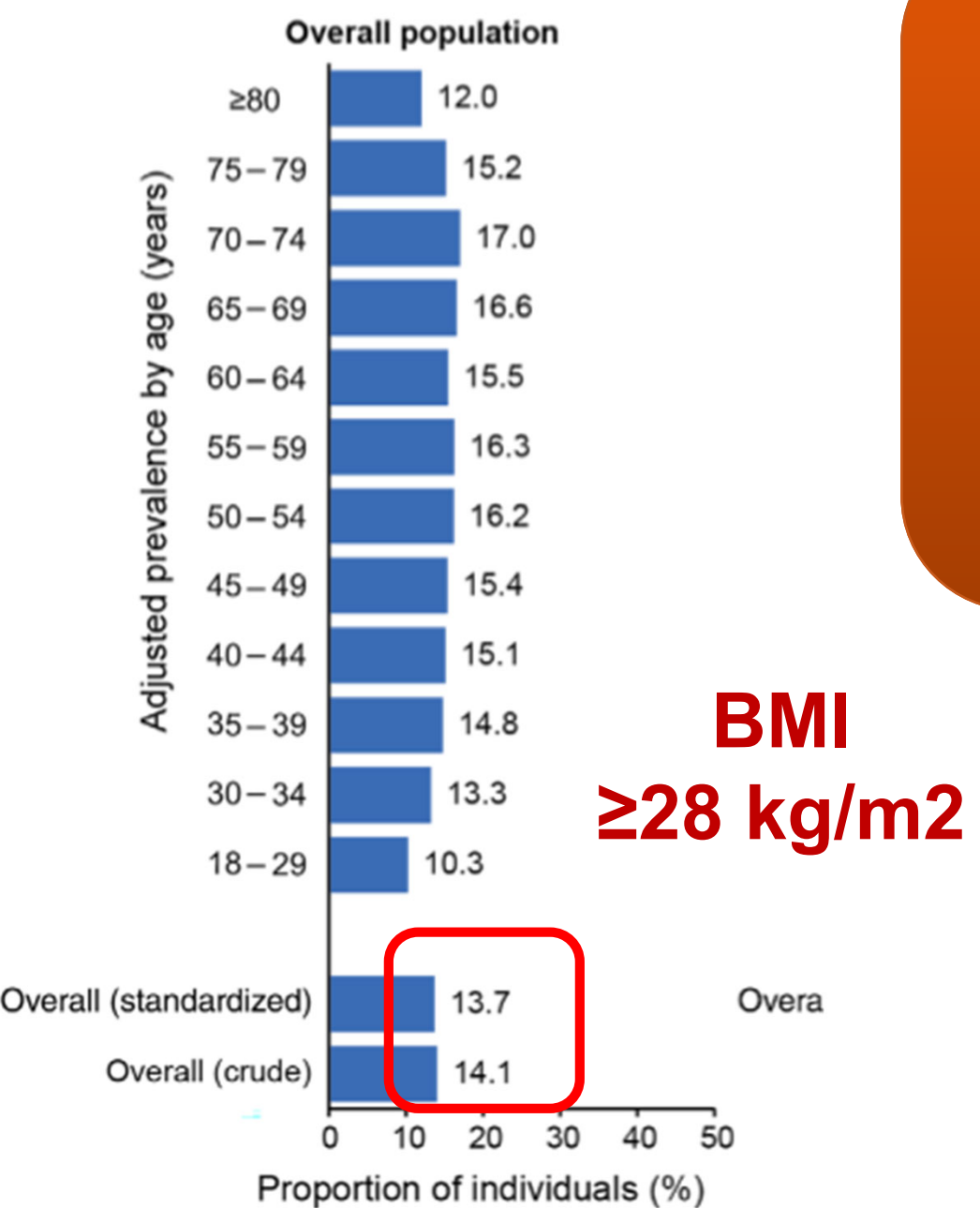
Obesity in China

Health screening statistics (n=21,7871,683)

(A) Prevalence of overweight



(B) Prevalence of obesity



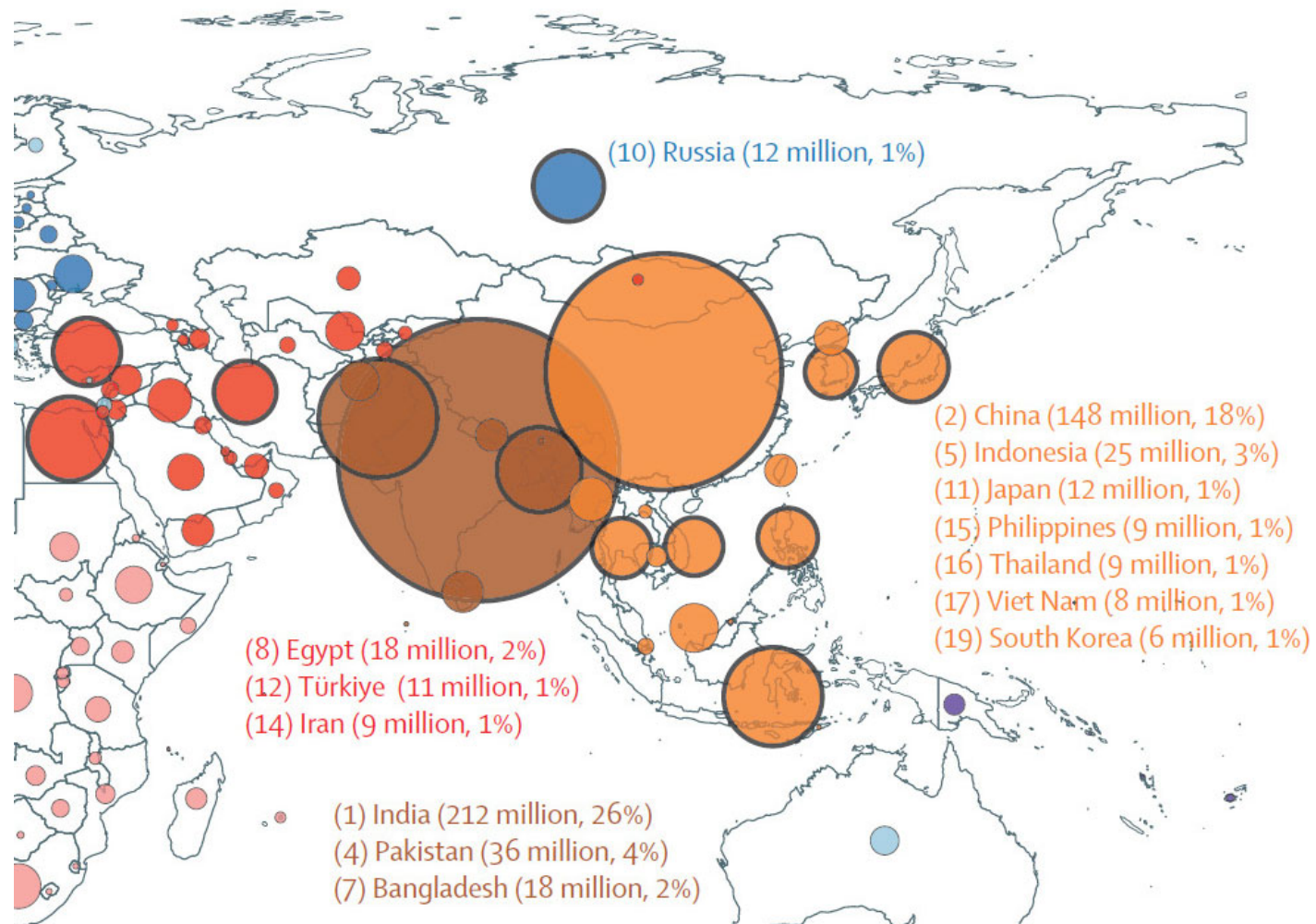
Obesity
Prevalence
2004 : 3.1%
2018 : 8.1%

Wang L et al. Lancet 2021

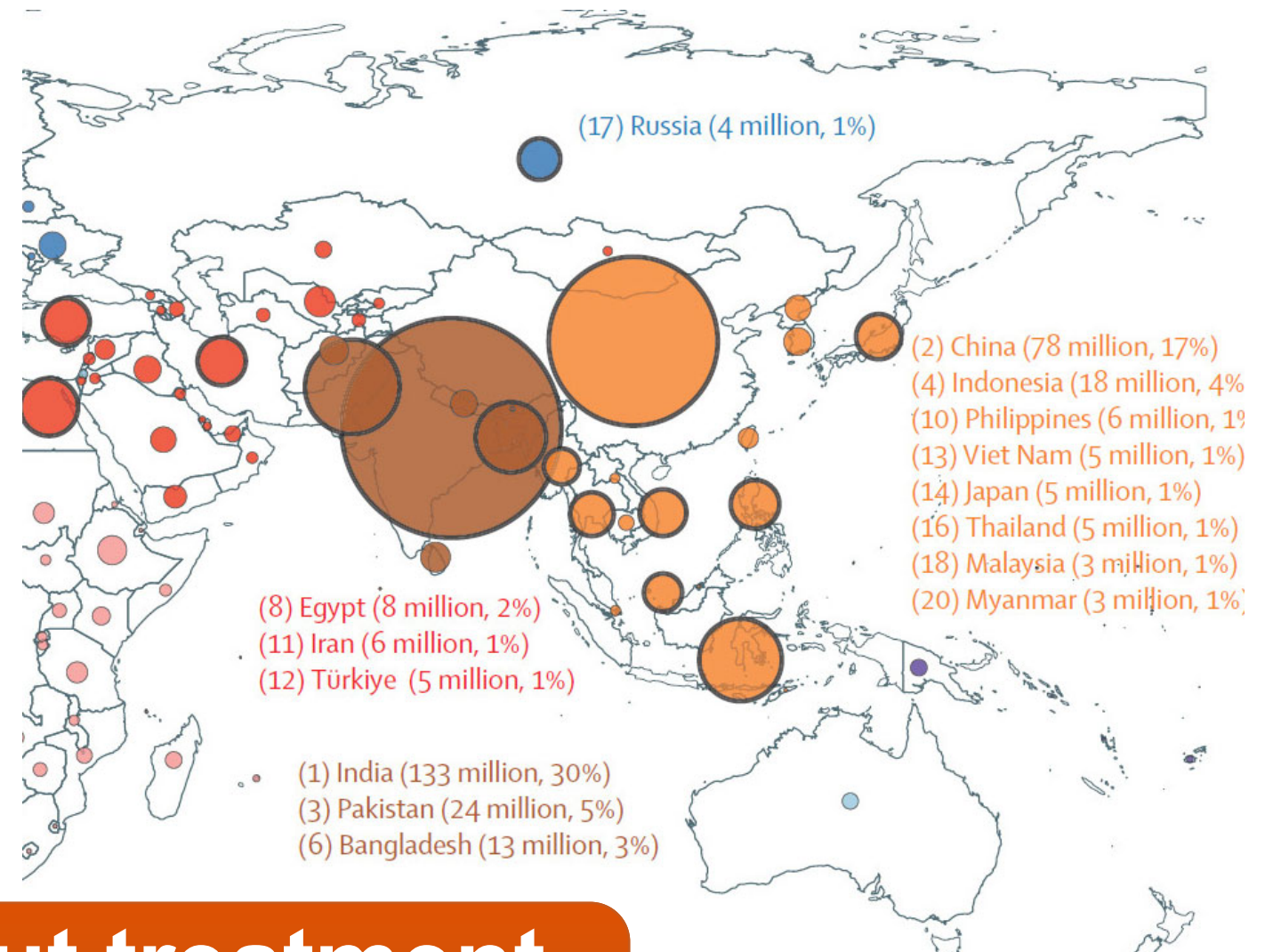
Diabetes in Asia

High prevalence, low treatment rates

Prevalence



Without Treatment



**China: Diabetes without treatment
52%**

Steatosis vs. Steatohepatitis

Steatosis (MASLD)

- Purely fat in liver
- Low risk of disease progression
- Can progress to MASH
- Dynamic changes with time
- Burn-out in late-stage

Steatohepatitis (MASH)

- Liver biopsy: inflammation and ballooning
- ↑ ALT/AST (low accuracy)
- Increased risk of disease progression
 - Fibrosis / Cirrhosis
 - HCC
- **No Good Non-invasive Biomarker**

MASLD in Primary Care

Key: exclude **Advanced Fibrosis**

Fibrosis-4 (FIB-4) Index for Liver Fibrosis

Noninvasive estimate of liver scarring in HCV and HBV patients, to assess need for biopsy.

When to Use ▾

Pearls/Pitfalls ▾

Why Use ▾

Age

Use with caution in patients <35 or >65 years old, as the score has been shown to be less reliable in these patients

65

years

AST

Aspartate aminotransferase

45

U/L

ALT

Alanine aminotransferase

65

U/L

Platelet count

145

$\times 10^3/\mu\text{L}$

2.50 points

Further investigation needed

Approximate fibrosis stage: Ishak 2-3 (Sterling et al 2006)

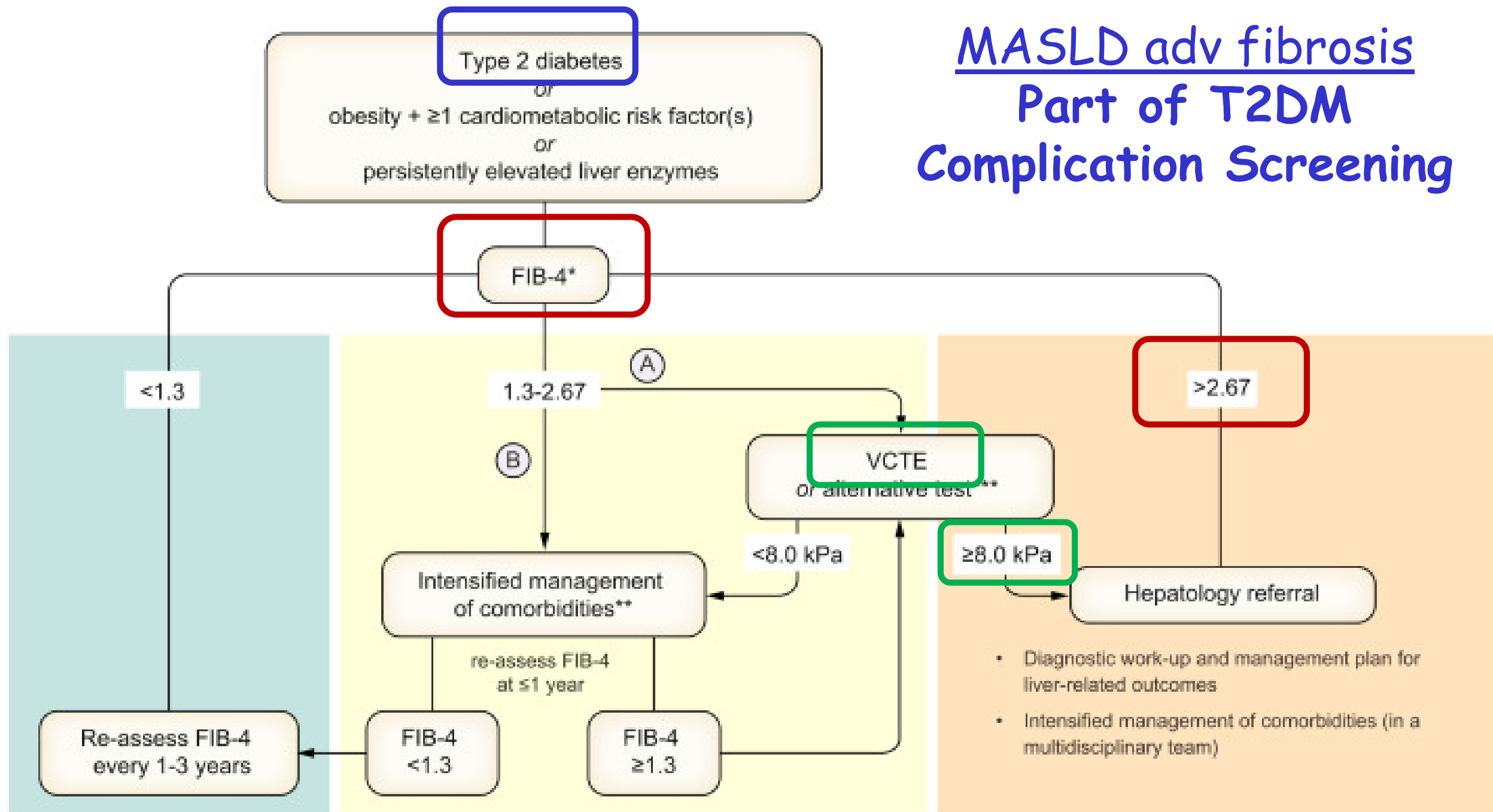
Copy Results 📄

Next Steps >>>

- **FIB-4: Exclude** advanced fibrosis
- Elastography: Detect adv. Fibrosis
— *(Optional)*



FIB-4: Most Important Test in MASLD



Additional Assessment in MASLD

Liver and non-Liver

- HCC surveillance for advanced fibrosis and cirrhosis

- Ultrasound* +/- AFP



- ?Liver biopsy
- ?MRI-PDFF

- Cardiometabolic

- T2DM / prediabetes (A1c, FBG etc)
 - Obesity or overweight
 - Dyslipidemia
 - Cardiovascular risk assessment

- OSA

- Kidney disease

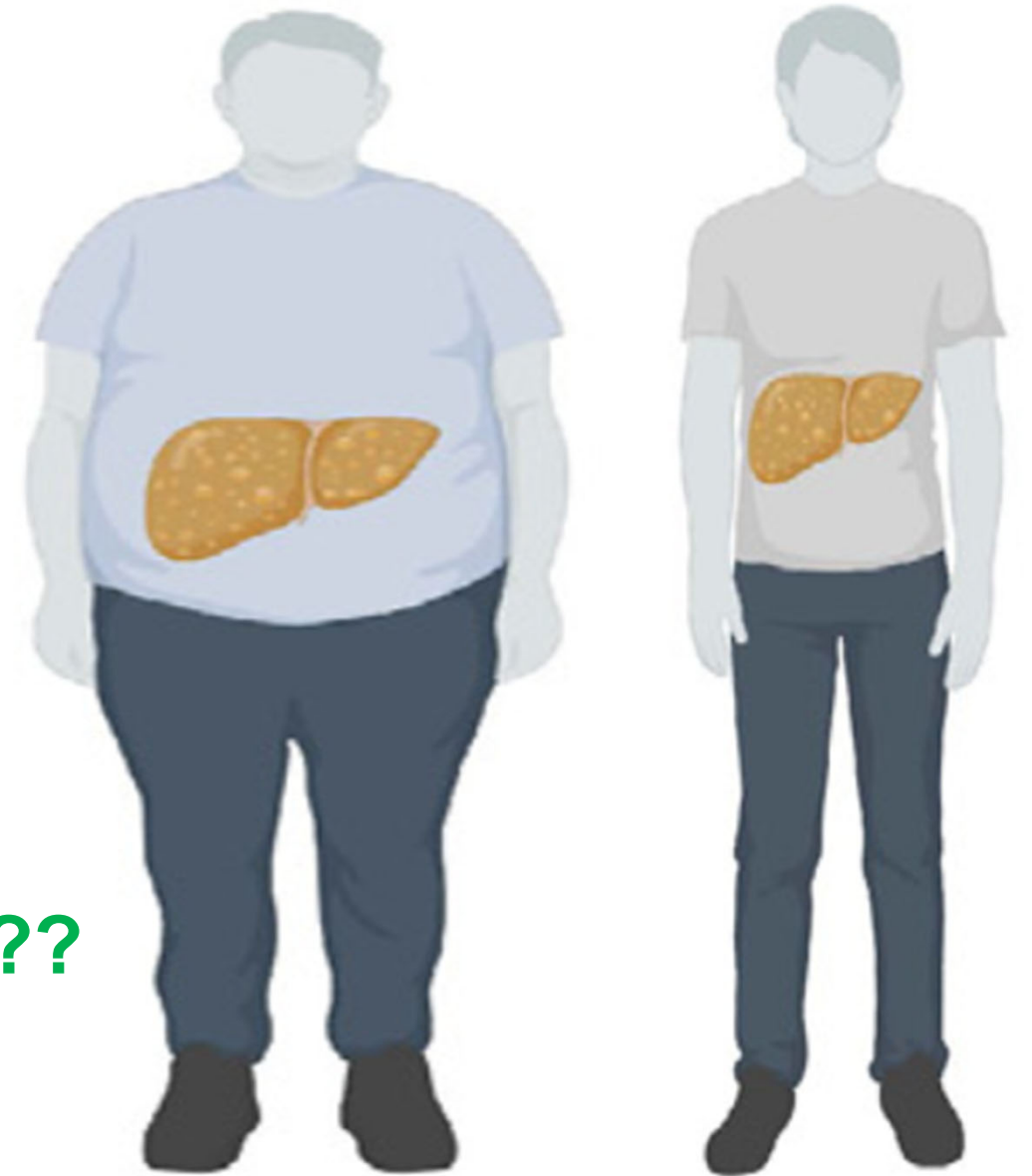
Non-Pharmacological Management

- **Weight loss Targets**

- Simple steatosis: 5%
- Inflammation ($\uparrow$ ALT): 7%
- Fibrosis: 10%
- Normal body weight: 3-5%

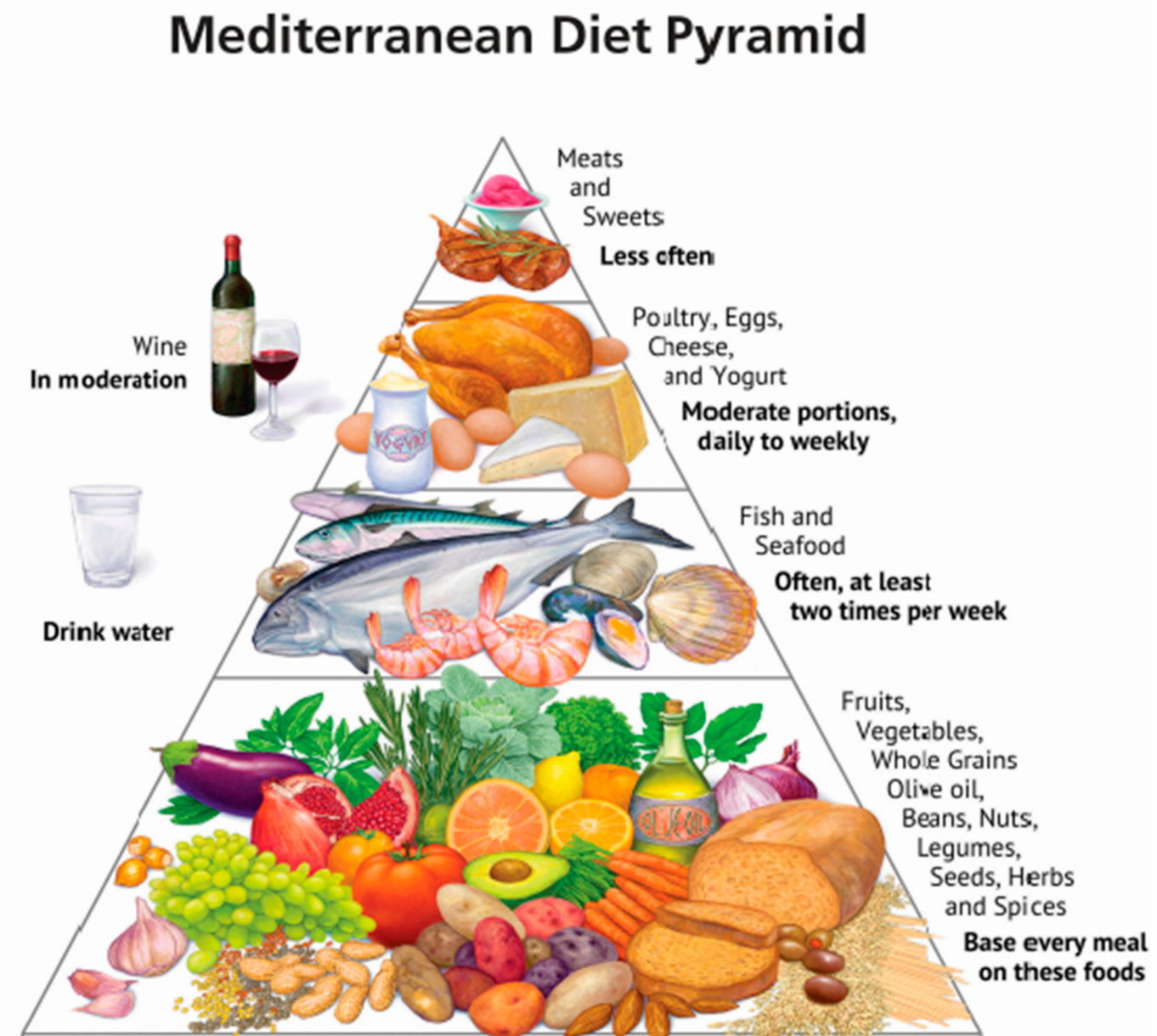
- Dietary strategies: need **individualization**

- Hypocaloric diet ($<1,000$ kcal/day)
 - Ketogenic diet (Carb: $<20-50$ g/day)
 - ?? Time-restricted eating
- } **Sustainable??**



Non-Pharmacological Management

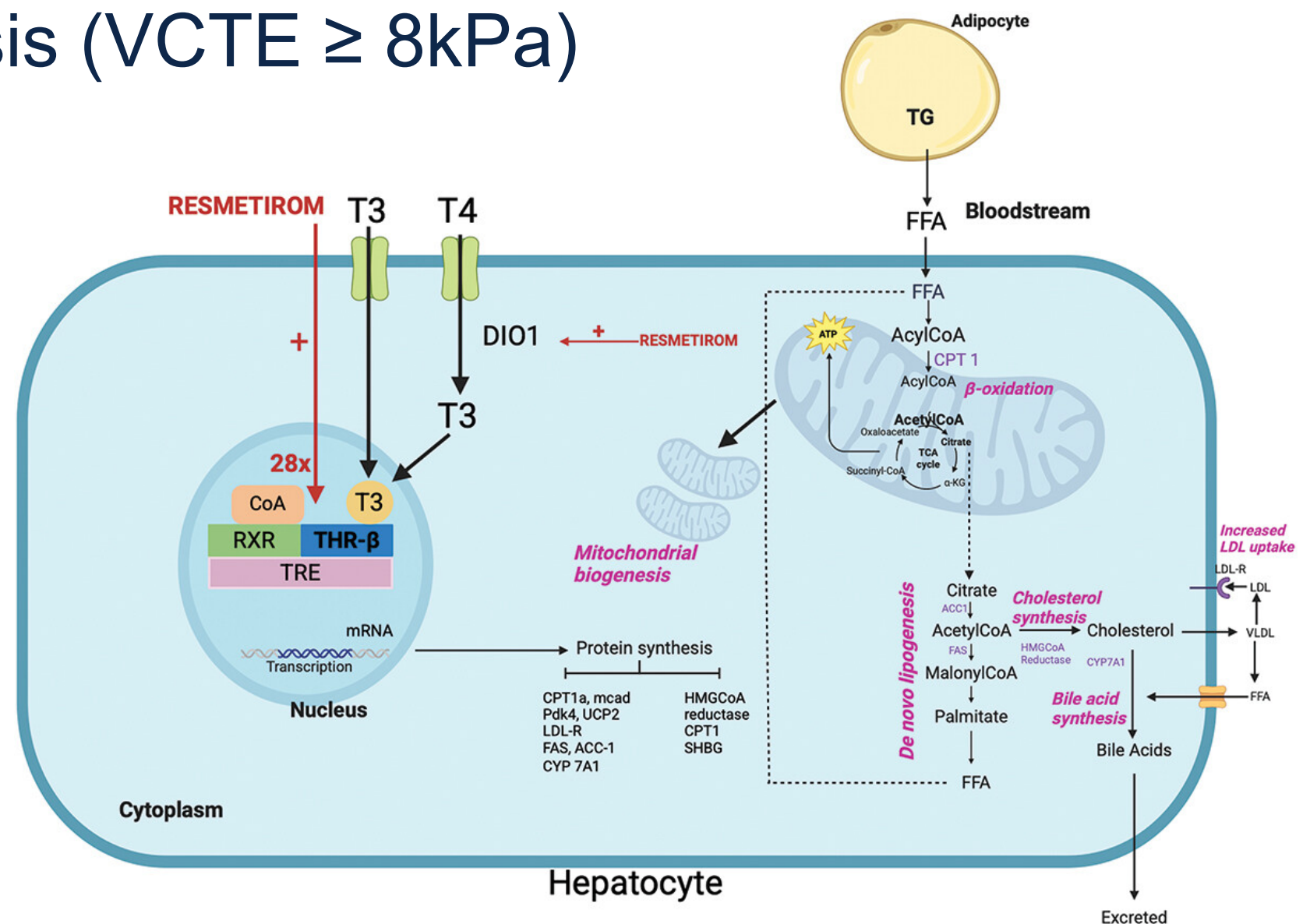
- Other Dietary Advice:
 - Mediterranean diet pattern
 - ↓ **Ultra-processed** foods (red meat)
 - ↓ ↓ **Sugar-sweetened** beverages (fructose)
 - Coffee!! (≥ 3 cups per day) 🤔 🤔
- Exercise
 - >150 min/week mod intensity or
 - >75 min/week high intensity
 - Reduce sedentary time
- Reduce smoking / alcohol



Pharmacological Treatment: Resmetirom

- Liver-directed thyroid hormone receptor (THR) agonist
- Suspected MASH with fibrosis (VCTE ≥ 8 kPa)
- Not licensed for cirrhosis
- Not available in Asia

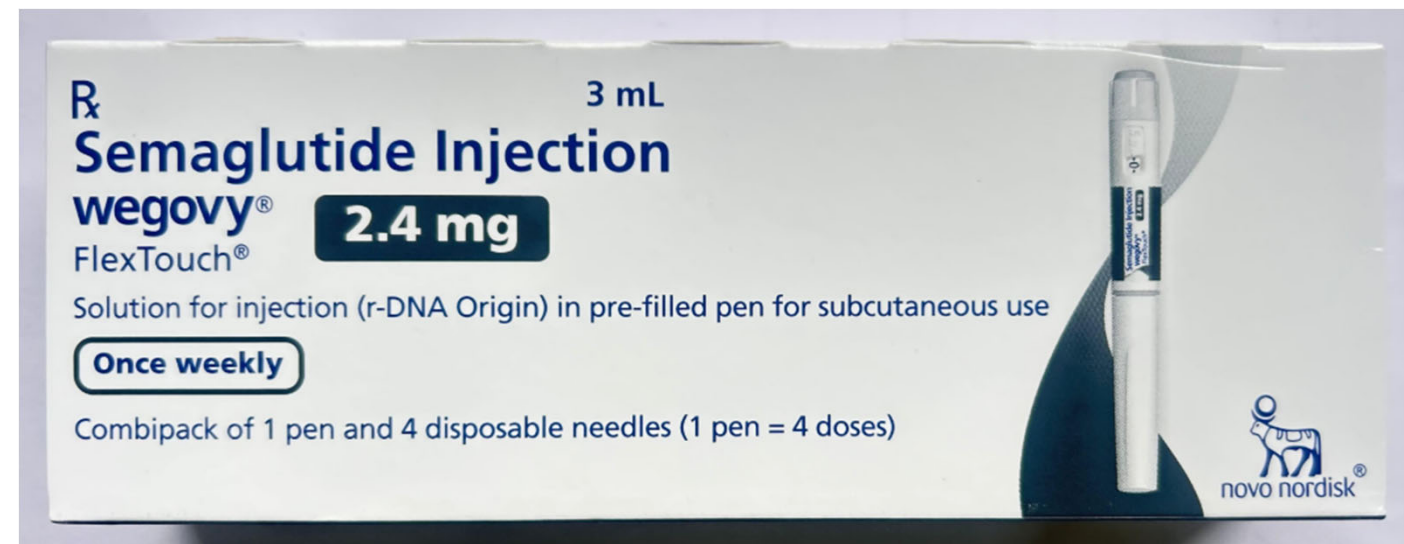
Modest effect
25-30% with **MASH resolution**
and **fibrosis improvement**



Pharmacological Treatment: GLP-1A

- Semaglutide (Glucagon-like peptide-1 receptor agonists)
- **Body weight / fat loss**
 - Suppress glucagon / activate insulin
 - ↓ **gastric emptying** / appetite
 - 36-62% MASH treatment response
 - *[Western pop: BMI 34 kg/m²]*
 - Mean weight loss >10%
 - Need gradual dose escalation
 - Oral version available in future

Indication:
Suspected MASH with fibrosis
(VCTE ≥8kPa)



Multi-system effect of GLP-1A

Good? Bad?

• GI side effects

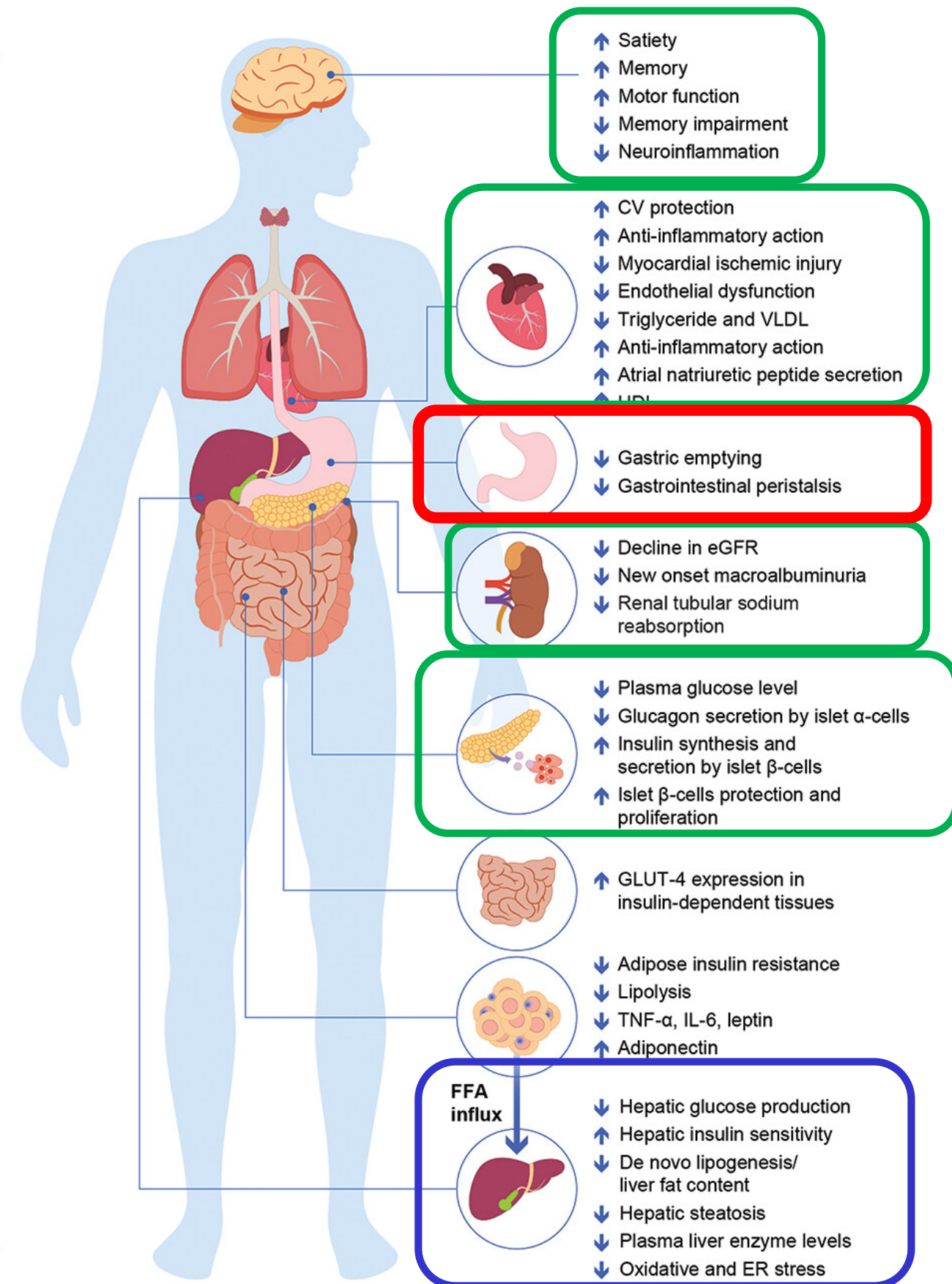
- **Gastroparesis** / Vomiting / Nausea
- Others (constipation, pain)
- Need gradual dose escalation

- Pancreatitis* (older versions mainly)
- Gallstone* (reduced GB motility)

• Off-Label Use



"Ozempic Face"



Other Therapies for MASLD / MASH

- ?Vit E (? Increased all-caused mortality)
- Safe to use in MASLD:
 - Statins
 - Metformin
 - SGLT2 inhibitors
- Treating co-morbidities also important!
- Bariatric endoscopy / surgery

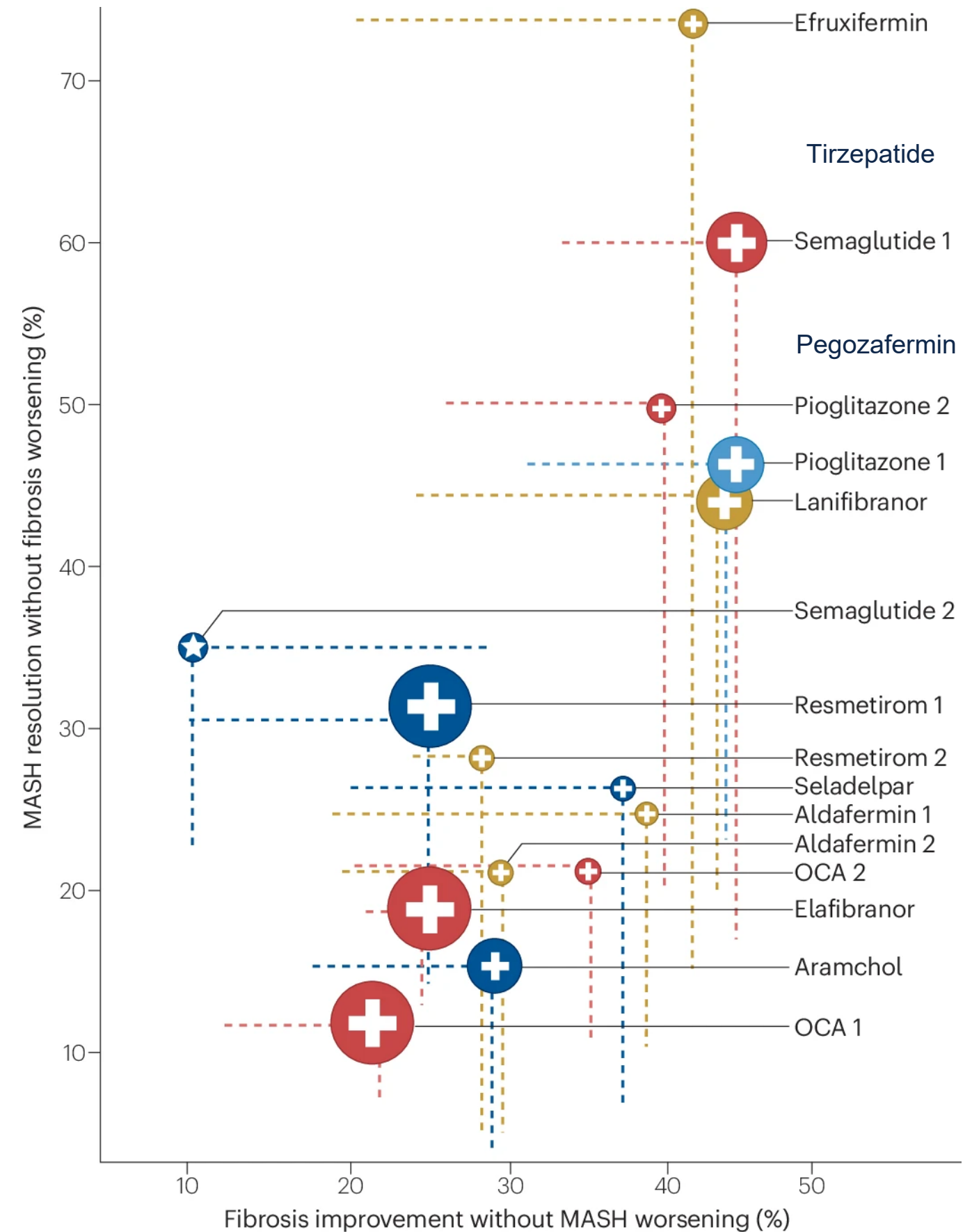
**WHAT'S
NEXT?**

Tirzepatide

Oral GLP-1A

PPAR Agonist

FGF21 Analogue



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- Alcohol and the Liver
 - Alcoholic fatty liver
 - Alcoholic hepatitis
- Metabolic associated steatotic liver disease (MASLD)
 - MASLD and MASH
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Related WCSs

- **Cirrhosis & ascites**
- **Liver failure**
- **I am overweight**
- **Alcohol & the brain**

Thank You!!



Prof. Wai-Kay Seto

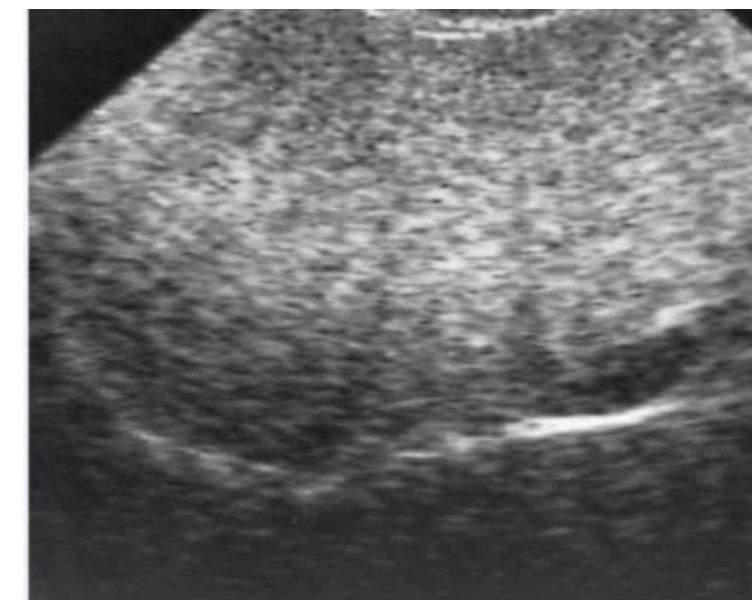
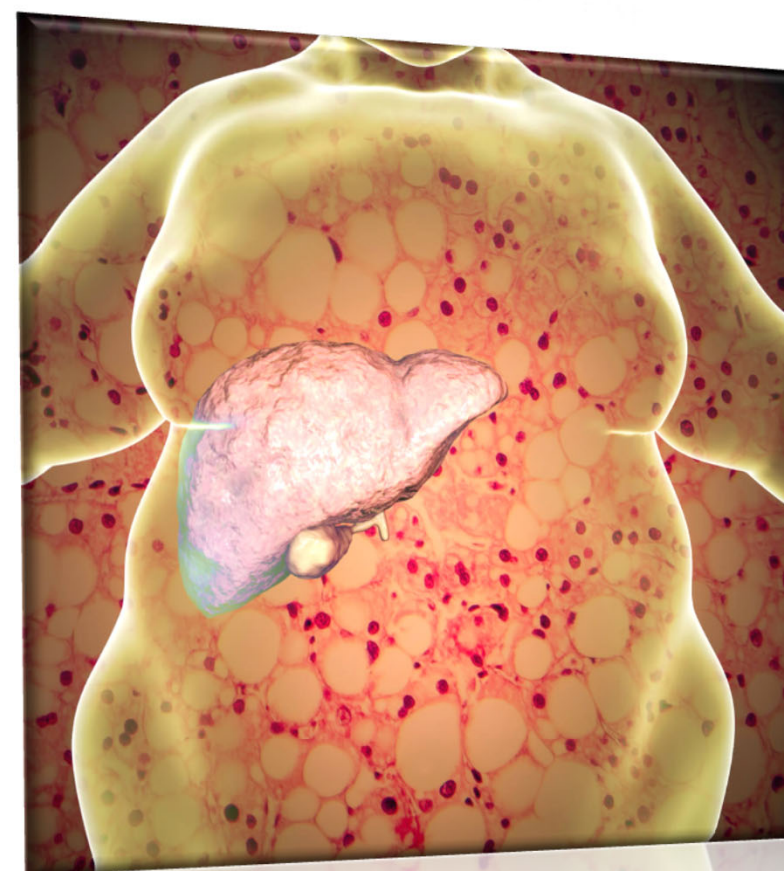
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**HKU
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School of Clinical Medicine
Department of Medicine
香港大學內科學系



TELL US WHAT

GC Whole Class Session

YOU THINK

THANK

YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)